# Studying the Sustainability of AI Generated Code

Chelsea Wiredu | chelsea25@vt.edu  Chris Brown | dcbrown@vt.edu

Department of Computer Science at Virginia Tech

## INTRODUCTION

- AI-generated software is growing in popularity, leading to an increasing environmental impact.
- I used CodeCarbon to measure and compare lifecycle carbon emissions between AI models and human-written code.
- This study examines how prompt complexity scales emissions and compares the cumulative carbon footprint of both software types.

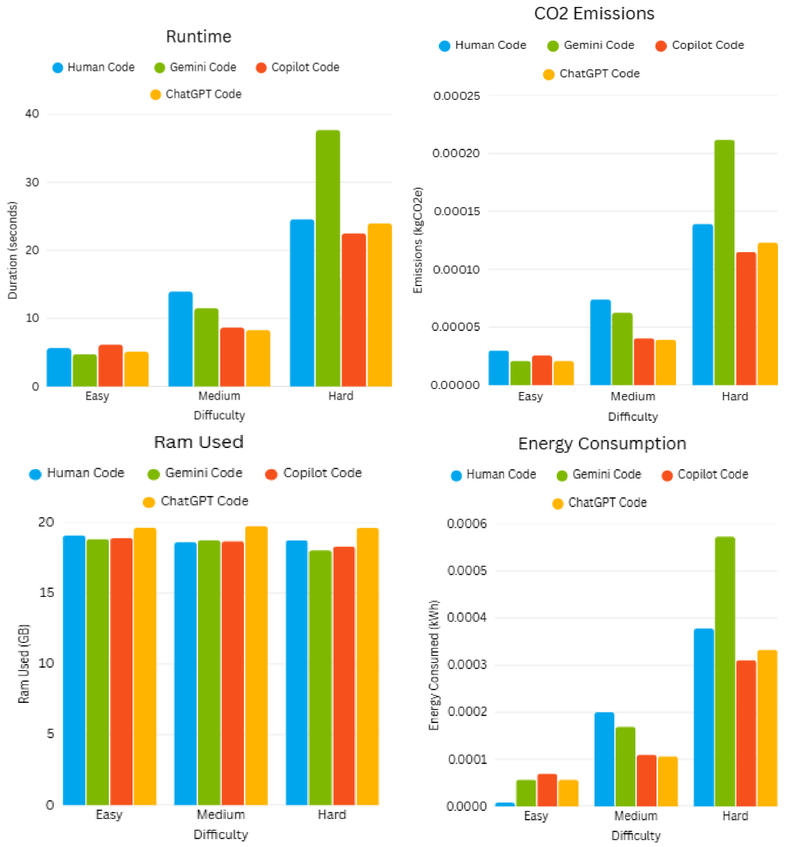
## RESEARCH QUESTIONS

1. How do the carbon emissions and energy consumption of LLM code generation change as prompt complexity increases?
2. How does the overall carbon footprint of generating and running AI code compare to executing human-authored code?

## METHODS

- This study compares code solutions generated by Gemini (3.5 Flash), ChatGPT (GPT-5.5), and Copilot (Free) against human-written code.
- All solutions were tested using CodeCarbon to track CO2 emissions, energy usage, runtime, and RAM usage.
- The analysis compares the average duration, emissions, energy usage, and RAM usage across each category.

## RESULTS




| project_name | duration | emissions | energy_consumed | ram_used_gb |
|---|---|---|---|---|
| _human_sol_EASY_Q1_R1 | 5.971453 | 3.73E-05 | 0.000101 | 19.42347 |
| _human_sol_EASY_Q1_R2 | 6.8173442 | 3.27E-05 | 8.87E-05 | 19.20702515 |
| _human_sol_EASY_Q1_R3 | 4.1886738 | 1.93E-05 | 5.22E-05 | 18.68575096 |
| _human_sol_MED_Q1_R1 | 17.723635 | 8.74E-05 | 0.000236884 | 18.69517059 |
| _human_sol_MED_Q1_R2 | 6.0182944 | 3.15E-05 | 8.53E-05 | 18.33640289 |
| _human_sol_MED_Q1_R3 | 18.1429274 | 0.000102636 | 0.000278046 | 18.81707644 |
| _human_sol_HARD_Q1_R1 | 10.0977105 | 5.34E-05 | 0.000144735 | 19.27169037 |
| _human_sol_HARD_Q1_R2 | 25.49697 | 0.000154669 | 0.000419004 | 18.82260994 |
| _human_sol_HARD_Q1_R3 | 38.1187846 | 0.000210335 | 0.000569806 | 18.1593702 |
| _Gemini_Easy_Q1_R1 | 3.9266076 | 1.86E-05 | 5.05E-05 | 19.13191605 |
| _Gemini_Easy_Q1_R2 | 5.606612 | 2.24E-05 | 6.08E-05 | 18.67044957 |
| _Gemini_Easy_Q1_R3 | 4.7469265 | 2.20E-05 | 5.95E-05 | 18.72180176 |
| _Gemini_Med_Q1_R1 | 11.3885915 | 5.44E-05 | 0.000147458 | 18.77349854 |
| _Gemini_Med_Q1_R2 | 12.25468 | 7.02E-05 | 0.000190252 | 18.8119072 |
| _Gemini_Med_Q1_R3 | 10.9005417 | 6.30E-05 | 0.000170723 | 18.69607925 |
| _Gemini_Hard_Q1_R1 | 30.7160309 | 0.000172509 | 0.000467335 | 18.26134627 |
| _Gemini_Hard_Q1_R2 | 25.7417945 | 0.000142441 | 0.000385878 | 18.04064525 |
| _Gemini_Hard_Q1_R3 | 56.6042467 | 0.000319793 | 0.000866331 | 17.80494726 |

## RESULTS CONT.

| project_name | duration | emissions | energy_consumed | ram_used_gb |
|---|---|---|---|---|
| _Copilot_EASY_Q1_R1 | 4.313541 | 1.62E-05 | 4.39E-05 | 18.80833244 |
| _Copilot_EASY_Q1_R2 | 5.8522835 | 2.22E-05 | 6.01E-05 | 18.95267995 |
| _Copilot_EASY_Q1_R3 | 8.3702953 | 3.86E-05 | 0.000104567 | 18.97373962 |
| _Copilot_MED_Q1_R1 | 8.5076909 | 3.91E-05 | 0.00010602 | 18.55116425 |
| _Copilot_MED_Q1_R2 | 8.2039621 | 3.81E-05 | 0.00010325 | 18.80009766 |
| _Copilot_MED_Q1_R3 | 9.2740213 | 4.42E-05 | 0.000119826 | 18.68654188 |
| _Copilot_HARD_Q1_R1 | 36.1313048 | 0.000189046 | 0.000512134 | 18.3522526 |
| _Copilot_HARD_Q1_R2 | 22.8990814 | 0.00011716 | 0.000317391 | 18.39551582 |
| _Copilot_HARD_Q1_R3 | 8.5227355 | 3.83E-05 | 0.000103626 | 18.17190018 |
| _Chat_EASY_Q1_R1 | 4.1742682 | 1.56E-05 | 4.24E-05 | 19.57997131 |
| _Chat_EASY_Q1_R2 | 5.4147782 | 2.39E-05 | 6.48E-05 | 19.72858683 |
| _Chat_EASY_Q1_R3 | 5.7944155 | 2.29E-05 | 6.20E-05 | 19.63007991 |
| _Chat_MED_Q1_R1 | 7.3593243 | 3.32E-05 | 8.98E-05 | 19.6945015 |
| _Chat_MED_Q1_R2 | 9.1161299 | 4.29E-05 | 0.00011619 | 19.91607602 |
| _Chat_MED_Q1_R3 | 8.4678817 | 4.18E-05 | 0.000113294 | 19.68923505 |
| _Chat_HARD_Q1_R1 | 19.3496215 | 9.45E-05 | 0.000255917 | 19.90595126 |
| _Chat_HARD_Q1_R2 | 28.4285211 | 0.000150075 | 0.000406559 | 19.63433914 |
| _Chat_HARD_Q1_R3 | 24.1239055 | 0.00012452 | 0.000337331 | 19.36850433 |

- Copilot (0.000163 kWh | 12.45s): Most sustainable
- ChatGPT (0.000165 kWh | 12.46s): Second most sustainable.
- Human Solutions (0.000219 kWh | 14.73s): Third most sustainable.
- Gemini (0.000267 kWh | 17.98s): Least sustainable.

## CONCLUSION

- Longer runtime leads to higher energy usage and higher CO2 emissions
- Lower RAM usage does not guarantee increased sustainability if the duration of the program is longer

### ACKNOWLEDGEMENTS

- Mentorship & Academic Outreach Program
- The Department of Computer Science in the College of Engineering